

Dimensional Data Modelling Design for Palladium Bank Retail Banking Analytics

Palladium Bank's existing reporting environment currently relies on raw transaction logs, which limits reporting speed, consistency, and analytical flexibility. To address this challenge, a dimensional data warehouse model was designed using a star schema approach to support customer behavior analytics, branch performance reporting, product monitoring, churn identification, and time-based comparisons. The proposed model organizes transactional data into one central fact table linked to multiple conformed dimensions for efficient querying and scalable reporting.

The proposed dimensional model was intentionally structured to answer Palladium Bank's core retail analytics questions. Customer fee-generating segments can be analyzed through the linkage between Fact_Transactions, Dim_Customer, and transaction amount metrics segmented by Tier. Transaction behaviour variation across branches and channels is enabled through Dim_Branch and Dim_Channel relationships. Potential reduction in activity among high-value customers can be tracked through recurring transaction frequency, recency, and balance movement across the date and customer dimensions. In addition, Dim_Product allows management to distinguish which banking products are associated with stronger deposit flows versus withdrawal activity. The inclusion of the Date dimension further supports month-on-month, quarter-on-quarter, and year-on-year trend comparisons across all analytical categories.

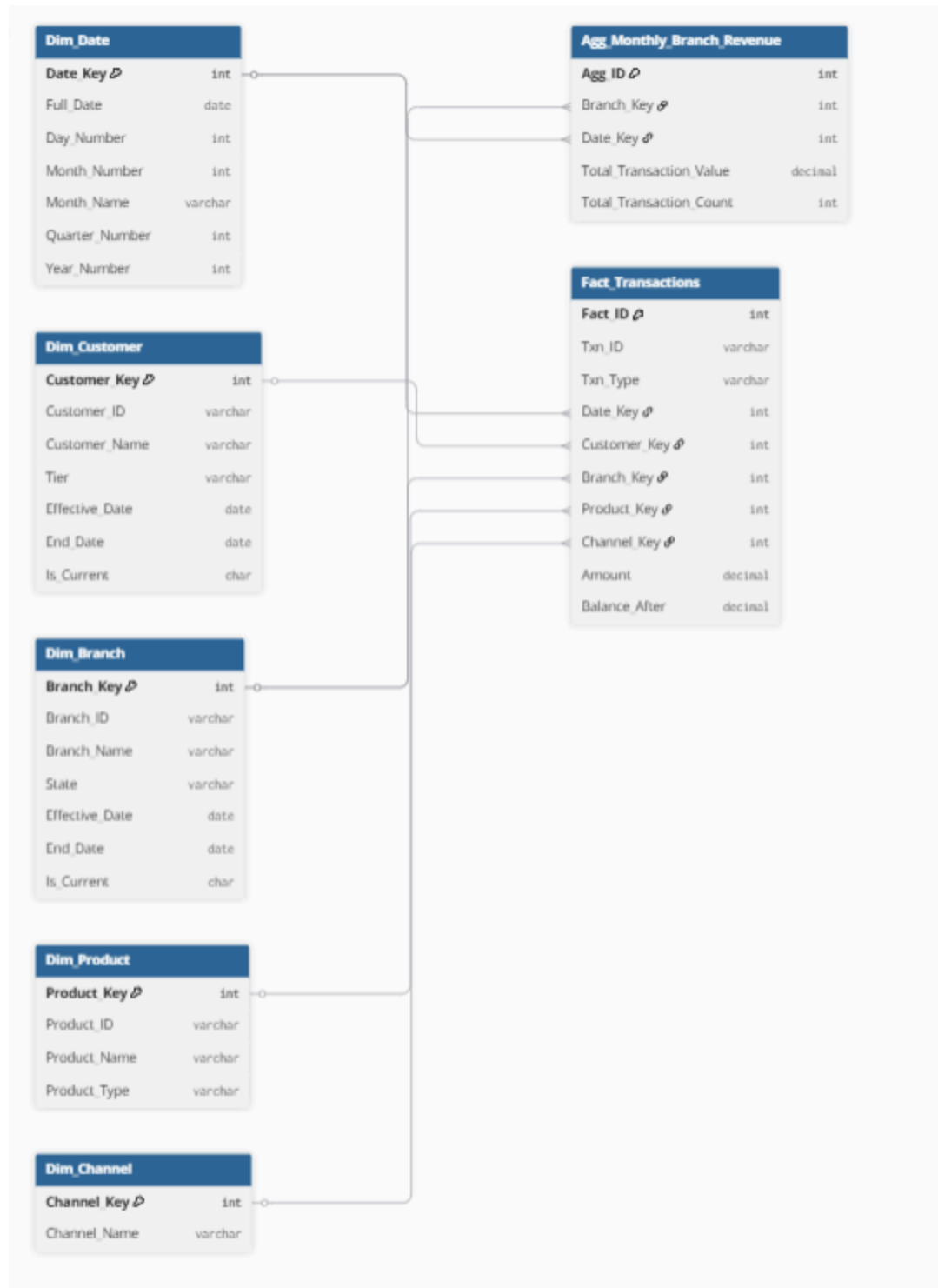


Figure 1: Palladium Bank Retail Banking Star Schema Design

1. Fact Table Grain and Business Process

The selected business process for Palladium Bank is retail customer transaction monitoring, since all required business questions are centred on customer deposits, withdrawals, channel usage, branch performance, and product interaction. The grain of the fact table was defined at the individual transaction level, meaning that each row in the Fact_Transactions table represents one unique customer banking transaction executed at a particular date, branch, product, and service channel. This granular level was selected because it provides the flexibility required for detailed customer behaviour analysis, trend comparisons, fee income monitoring, churn signal detection, and branch-level performance reporting without losing transactional precision.

2. Dimension Identification and Degenerate Dimensions

Based on the provided source dataset, five conformed dimensions were identified to support descriptive slicing of the transaction facts. Dim_Date was created from the transaction date field to enable month-on-month, quarter-on-quarter, and year-on-year reporting. Dim_Customer captures customer descriptive attributes such as Customer_ID, Customer_Name, and Tier for customer segmentation analysis. Dim_Branch stores Branch_ID, Branch_Name, and State to support geographic branch performance comparisons. Dim_Product stores Product_ID, Product_Name, and Product_Type to monitor which products drive deposits and withdrawals. Dim_Channel stores the transaction medium, such as Branch, POS, Mobile App, or Internet Banking, for service channel analysis. In addition, Txn_ID and Txn_Type were retained inside the fact table as degenerate dimensions because they describe each transaction event but do not require standalone descriptive tables.

Each dimension was assigned a surrogate primary key to create stable warehouse joins, simplify foreign key relationships, and preserve historical tracking independently from changing business identifiers in the source system.

3. Slowly Changing Dimension Strategy

Slowly Changing Dimension handling was necessary because some descriptive attributes may evolve over time, while historical reporting accuracy must be preserved.

SCD Type 2 was preferred over Type 1 because overwriting changing descriptive values would remove the historical context needed for long-term customer and branch trend analysis.

For this reason, Dim_Customer was designed using SCD Type 2. A customer may migrate from one service tier to another, for example, from Silver to Gold, and the bank still needs to know which tier generated historical transactions. Effective_Date, End_Date, and Is_Current fields were therefore included to maintain historical versions of customer records. Dim_Branch was also assigned SCD Type 2 because branch descriptors may change due to renaming, relocation, or administrative restructuring. This ensures that branch-based trend analysis remains historically consistent even after organizational changes.

4. ETL / ELT Loading Approach

For the initial historical load, the recommended process is to first populate all dimension tables before loading the fact table. Distinct dates are extracted into Dim_Date, followed by unique customer, branch, product, and channel records into their respective dimensions, where surrogate keys are generated automatically.

Required transformations during the historical load include duplicate transaction removal, date standardisation, extraction of distinct dimension members, surrogate key assignment, and replacement of source business identifiers with warehouse lookup keys.

After all dimensions are populated, Fact_Transactions is loaded by replacing business identifiers with their corresponding surrogate foreign keys. Incremental daily loads would then detect only new transaction records using Txn_ID validation, while changed customer or branch records would trigger SCD Type 2 inserts with old records expired and new records activated. Duplicate fact prevention is achieved by enforcing transaction ID uniqueness during each load cycle.

These checks are detected through null scans, uniqueness validation, date format validation, and referential integrity joins; any records failing validation are quarantined into an exception table for cleansing before warehouse insertion.

5. Data Quality, Performance, and Scalability

The model supports two primary analytical drill-down hierarchies: Year → Quarter → Month → Day through Dim_Date, and State → Branch through Dim_Branch for progressive geographic branch analysis.

To ensure warehouse reliability, several data quality controls are required, including null transaction ID detection, duplicate transaction detection, invalid date validation, and orphan foreign key checks where a fact record references a missing dimension member. Any failed records should be redirected to an exception log for review before final loading. For scalability, the Fact_Transactions table should be partitioned monthly because Palladium Bank processes large time-series transaction data across multiple cities. Foreign key indexes were also created on all dimension joins to accelerate query performance. In addition, an aggregation table named Agg_Monthly_Branch_Revenue was introduced to pre-summarise monthly transaction value and transaction count by branch, thereby reducing report runtime for management dashboards and recurring executive analysis.

In summary, the star schema architecture was selected because it provides a simplified and query-efficient analytical structure compared to reporting directly from raw operational transaction logs. By separating descriptive attributes into conformed dimensions and storing numeric transactional activity in a central fact table, Palladium Bank gains faster reporting, better consistency, easier drill-down analysis, and future scalability as transaction volume grows. The inclusion of surrogate keys, SCD tracking, indexed foreign keys, and pre-aggregated summaries further ensures that the warehouse can support both detailed operational analytics and high-level executive reporting requirements.

Source Column	Classification
Txn_ID	Degenerate Dimension

Txn_Date	Dimension (Date)
Customer_ID	Dimension (Customer)
Customer_Name	Dimension (Customer)
Tier	Dimension (Customer)
Branch_ID	Dimension (Branch)
Branch_Name	Dimension (Branch)
State	Dimension (Branch)
Product_ID	Dimension (Product)
Product_Name	Dimension (Product)
Product_Type	Dimension (Product)
Txn_Type	Degenerate Dimension
Channel	Dimension (Channel)
Amount	Fact Measure
Balance_After	Fact Measure

Table 1: Source-to-Dimensional Classification of Palladium Bank Transaction Fields